

AMC 10 Crash Course • Test T3 • Number Theory

10 problems • 30 minutes • no calculator • mastery bar 7 of 10

Name: _____

Date: _____

1. How many positive divisors does 540 have?
(A) 20 (B) 24 (C) 27 (D) 30 (E) 36
2. How many positive integers not exceeding 100 have exactly 12 divisors?
(A) 3 (B) 4 (C) 5 (D) 6 (E) 7
3. What is $\gcd(2024, 748)$?
(A) 4 (B) 11 (C) 22 (D) 44 (E) 88
4. Two lighthouses flash: one every 8 seconds, the other every 14. They just flashed together. In how many seconds will that happen again?
(A) 14 (B) 22 (C) 28 (D) 42 (E) 56
5. Two numbers have $\gcd$ 4 and lcm 120; one of them is 24. What is the other?
(A) 10 (B) 16 (C) 20 (D) 24 (E) 30
6. What is the remainder when 5^{2026} is divided by 7?
(A) 0 (B) 1 (C) 2 (D) 4 (E) 6
7. What is the last digit of 7^{2023} ?
(A) 1 (B) 3 (C) 5 (D) 7 (E) 9
8. What are the last two digits of 11^{25} ?
(A) 01 (B) 21 (C) 31 (D) 51 (E) 71
9. How many trailing zeros does $60!$ have?
(A) 14 (B) 15 (C) 16 (D) 18 (E) 20
10. What is the exponent of 7 in the prime factorization of $100!$?
(A) 9 (B) 12 (C) 14 (D) 15 (E) 16

Answer Key and Review

Parent page: cut off or do not print for the student. Map misses to lessons: 1-2 → 3.1, 3-5 → 3.2, 6-8 → 3.3, 9-10 → 3.4.

1	2	3	4	5	6	7	8	9	10
B	C	D	E	C	C	B	D	A	E

1. $540 = 2^2 \cdot 3^3 \cdot 5$; $(2+1)(3+1)(1+1) = 24$.
2. $12 = 12 \cdot 1 = 6 \cdot 2 = 4 \cdot 3 = 3 \cdot 2 \cdot 2$: shapes p^5q, p^3q^2, p^2qr . Up to 100 this gives 60, 72, 84, 90, 96 — exactly 5.
3. Euclid: $2024 = 2 \cdot 748 + 528$; $748 = 528 + 220$; $528 = 2 \cdot 220 + 88$; $220 = 2 \cdot 88 + 44$; $88 = 2 \cdot 44$. Gcd = 44.
4. $\text{Lcm}(8, 14) = 8 \cdot 14 / 2 = 56$.
5. Product = $4 \cdot 120 = 480$; the other number is $480 / 24 = 20$.
6. The cycle of 5 mod 7 has length 6; $2026 \equiv 4 \pmod{6}$, $5^4 = 625 = 89 \cdot 7 + 2$: remainder 2.
7. Cycle 7, 9, 3, 1; $2023 \equiv 3 \pmod{4}$: digit 3.
8. $11^n \equiv 10n + 1 \pmod{100}$: $250 + 1 \equiv 51$.
9. $\lfloor 60/5 \rfloor + \lfloor 60/25 \rfloor = 12 + 2 = 14$.
10. $\lfloor 100/7 \rfloor + \lfloor 100/49 \rfloor = 14 + 2 = 16$.